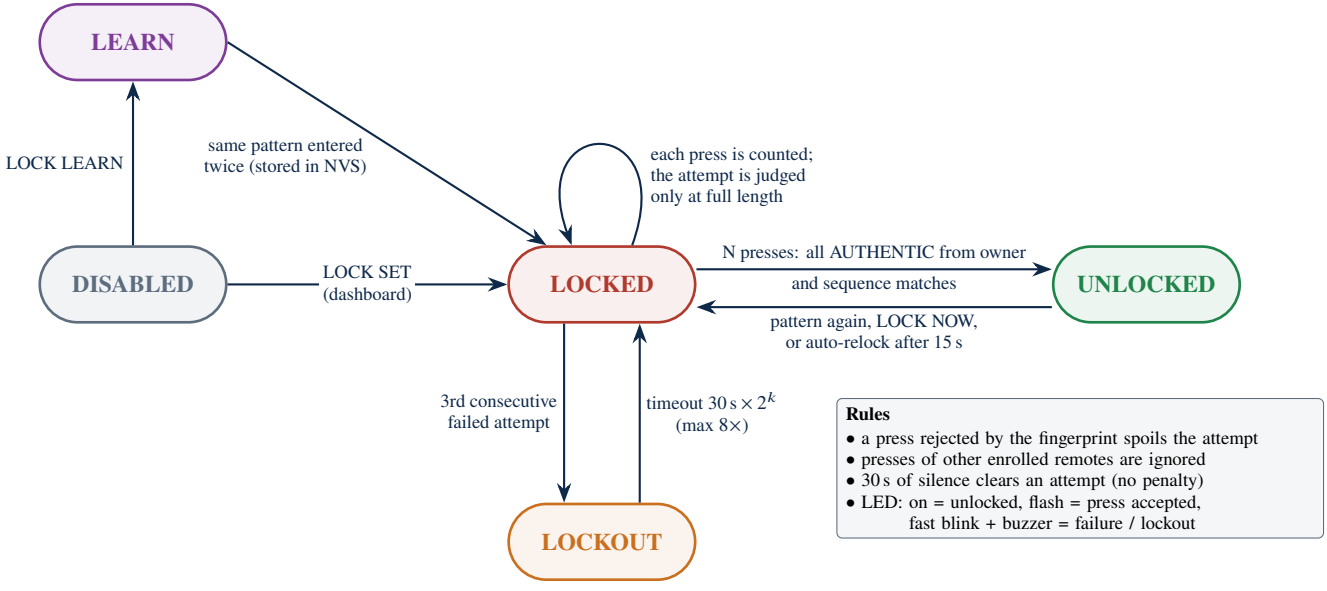

LEARN
LOCK LEARN
same pattern entered
twice (stored in NVS)
each press is counted;
the attempt is judged
only at full length
DISABLED
LOCK SET
(dashboard)
LOCKED
N presses: all AUTHENTIC from owner
and sequence matches
UNLOCKED
pattern again, LOCK NOW,
or auto-relock after 15 s
3rd consecutive
failed attempt
timeout 30 s × 2^k
(max 8×)
LOCKOUT
Rules
• a press rejected by the fingerprint spoils the attempt
• presses of other enrolled remotes are ignored
• 30 s of silence clears an attempt (no penalty)
• LED: on = unlocked, flash = press accepted,
fast blink + buzzer = failure / lockout